

Mode of Examination: Online
M.Sc. Semester-II Examination, 2021

2021

Subject: Computer Science

Paper Code & Name: CSM203 & (OBJECT ORIENTED SYSTEM)

Full Marks: 70

Date: 13.08.2021

Time: 3:00 Hours

Duration: 12:00 PM to 3:00 PM

Please note the following instructions carefully.

- **Promise not to commit any academic dishonesty.**
 - **Candidates are required to answer in their own words as far as applicable.**
 - **Each page of answer scripts should have your examination Roll Number on the right-side top corner of your answer script**
 - **The name of the scanned copy of the assessment script will be in the following format**
 - **(Example: CSM203-OOP-C91-CSC-191001.pdf)**
 - **The subject of the mail should be the assessment-script file name.**
 - **The scanned answer script is to be sent to cucse2020@gmail.com**
 - **Write the answers with a black ink ball pen.**
 - **The answer script should have an index page.**
 - **30 minutes is allotted for scanning and uploading the answer script.**
 - **Candidates are to check the readability and orientation of each scanned page of the answer script before uploading.**
 - **The Assessment script will be discarded if received after the specified date and time.**
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Answer Question No. 1, 2, and **any Four** from the rest

1. Answer any Five questions)

[2*5]

- a) “String class is immutable”- Illustrate with an example.
- b) Explain the major and minor elements of Object Modeling.
- c) Explain the IS-A relationship.
- d) What is the difference between an Exception and an Error?
- e) What is a garbage collector?
- f) “Object is said to have both state and behavior”- Explain.

2. Answer any Five questions:

[4 * 5]

- a) Consider the admission system in a college where a student gets admission by depositing admission fees at the account section, at the same time, he is registered in a hostel, academic section, and library system of the college. After completion of these procedures, he gets the identity card and the library card. Draw the corresponding activity diagram.
- b) What is Abstract Factory Pattern? What Design Patterns are used in the JDK library?

- c) What is the relation among JVM, JRE, and JDK? Why is the java platform an independent language?
- d) Define generalization & aggregation. Can we execute a java program without a main() method? Explain.
- e) What is the need for the Singleton pattern? How can we make a singleton class?.
- f) Can we overload the java main() method? Differentiate among final, finally, and finalize.

3. a) What is a marker or tagged interface? b) How can we implement multiple inheritances in Java? c) Why Dynamic method dispatch is called run-time polymorphism? Explain with suitable example. d) What is Varargs?

[1+3+4+2=10]

4. a) “The Abstract Factory pattern is to provide an interface for creating families of related or dependent objects without specifying their concrete classes” – Explain with example b) What is the difference between Abstract Factory and Factory pattern? c) What is the most common uses of the Adapter pattern? How does it help to deal with the interfaces of existing classes?

[2+2+(2+4) =10]

5. a) What is the difference between an Abstract class and interface? b) We can’t instantiate an abstract class, then why are constructors allowed in an abstract class? c) What is the difference between ‘throw’ and ‘throws’? Explain with example. d) What is an Anonymous object?

[2+2+4+2 =10]

6. a) What are the applications of Flyweight Pattern? Explain the structure & participants of the Flyweight pattern with a suitable example. What is the significance of the final block in Java exceptions?

[3+5+2 =10]

7. Short Note (Any Two)

[5 X 2 = 10]

- a. Model View Controller (MVC) design pattern.
- b. Associations.
- c. Wrapper Class.
- d. Checked and Unchecked Exception.